

PARTH SHAH

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EDUCATION

MS in Business Analytics and Artificial Intelligence, The University of Texas at Dallas Dec 2024

- **GPA** - 3.9, **Teaching Assistant** (Fall 23, Spring 24) - Econometrics and Time Series Analysis
- **Coursework** – ML, Deep Learning, Advanced Statistics, Programming for Data Science, Big Data, Analytics Platforms (AWS)

Bachelors in Computer Engineering, University of Mumbai May 2018

EXPERIENCE

Lead Product Analyst Jan 2022 - Jan 2023
Media.net *Mumbai, India*

- Led an **A/B test** to identify optimal request parameters for a DSP, resulting in 14% revenue increase. Partnered with PMs to design and analyze the experiment to provide insights for strategic decision-making, influencing **Product Roadmap**.
- Engineered a **Hadoop** data pipeline for consolidated advertiser performance metrics across 3 products from various data sources
 - Implemented **ETL process** aggregating 2 years of data (100+ TB) and hourly ingestion to **Apache Druid**.
 - Built a dashboard on Druid based Analytics platform, reducing report generation time from 2 days to instant access.
- Developed a POC for **XGBoost** model to optimize bid throttling's accuracy and recall, adding 5 new features. **Backtesting** demonstrated improved bid density; 12% reduction in blocked potential valid bids, and 1.5% revenue increase.
- Refactored **Hive SQL** queries to optimize 8 daily reports querying 3 TB+ databases by creating a summary table, significantly reducing execution time by 25% and resource consumption by 67%.
- Experimented with **sampling** strategies to reduce ad-request data logging from 10% to 4% and ensuring unbiased representation

Senior Product Data Analyst Jan 2021 - Dec 2021
Media.net *Mumbai, India*

- Engineered **predictive model** using **Spark SQL and Python** for auction win rates, incorporating real-time closing price feedback, resulting in a 3% increase in profit margins for a set of DSP and SSP.
- Led iterative **A/B testing** of 6+ native ad templates over 2 months; Achieved 3% CTR and 5% revenue improvement.
 - Worked with the PMs and engineers to log new parameters to analyze the test and UI/UX team to create native templates.
 - Set up, launched and monitored the A/B tests, analyzed the test results, and recommended the next steps.
- Investigated revenue discrepancy with a client and decreased it from 9% to 2.5% through **funnel and root cause analysis**. Deep-dived to identify attribution issue at browser version level; Collaborated with tech team and client to implement fixes.
- Optimized a heuristic service by refining 3 **SQL** queries, reducing bloom filter input data by 28% while maintaining accuracy.
- Conducted 20+ interviews to hire Analysts and trained them in SQL, Root Cause Analysis, debugging, and data analysis.

Product Data Analyst Jul 2018 - Dec 2020
Media.net *Mumbai, India*

- Collaborated with cross-functional teams to translate business requirements for programmatic products (Header Bidder, Prebid, Ad-Exchange) and resolved 2200+ discrepancies, ad-serving, reporting, and data logging tasks using **HiveQL / SQL, Python**.
- Conducted **EDA** and **statistical simulation in Python** to optimize ad expiry-time for an SSP, resulting in 6% uptick in ads.
- Devised real-time **KPI** dashboards and alert systems for programmatic products using internal BI tool (similar to **Tableau**), reducing issue response time to under 1 hour, enabling early detection of performance drops, and tracking \$50M+ in revenue.

PROJECTS

Predicting Hotel Booking Cancellations ([Link](#))

- Leveraged Sklearn's pipeline for data preprocessing and hyperparameter tuning of models (Random Forest, LightGBM, Logistic, XGBoost, CatBoost) using Random Search Cross validation and Bayesian Optimization to predict cancellations.
- Utilized stacked ensemble approach with a logistic regression meta-classifier to combine predictions from multiple base models to achieve 0.86 F1 score. Interpreted feature importance and explained model predictions using SHAP values.

LLM powered Streamlit App for scoring resume ([Link](#))

- Built dockerized Streamlit app to match resume and job description using Llama3-8B model hosted locally on Nvidia GPU.

Real-Time Data Pipeline for Stock Market Data | AWS EC2, Python, Kafka, Cassandra

- On an AWS EC2 instance, retrieved real-time stock data using Python, transformed and published it to a Kafka topic.
- Consumed the data from the Kafka topic and ingested it into a Cassandra Database.

Impact of Gun Laws on Crime Rate ([Link](#))

- Investigated the relationship between gun laws and crime rate by developing various Pooled OLS and panel data models (Entity Fixed Effect and Time Fixed Effect) and conducted hypothesis testing (using F-test) to assess the statistical significance.

SKILLS

Programming	SQL, Python (Pandas, NumPy, Matplotlib, scikit-learn, PyTorch, Tensorflow), R, HiveQL, SAS, STATA
Machine Learning	Regression, XGBoost, Random Forests, SHAP, Clustering, Time Series, PCA, Deep Learning, NLP, LLM
Big Data/Cloud	Hadoop, Hive, Spark, Kafka, Airflow, DBT, ETL pipeline, AWS, GCP, Snowflake, Databricks
Tools	Git, Docker, Tableau, Excel, Kibana, Mixpanel, Power BI, Alteryx, SQL Server, Google Ad Manager
Additional Skills	A/B Testing, Hypothesis Testing, REST APIs, Grafana, Linux, Elastic Search, Regex
Certifications	dbt Fundamentals, AWS Academy Data Engineering , Tableau for Data Scientists , Apache Airflow Essentials